WORLD FIRST SCIENTIFICALLY PROVEN ACTIVE ODOUR CONTROL + SOLAR-HYBRID ORGANIC WASTE PROCESSING UNIT

DRAVYA

Wet waste gets processed in 24hours — heated by the sun, sealed against odour, built in stainless steel.



Pre-production design visualisation · 500 kg/day unit shown

100 – 1,000+ kg/day
seven capacities

24 h cycle
batch-continuous

4-stage
odour train

SS304
wetted parts

Up to 40%
heater energy from the sun

01 Feed, any organic waste

Mixed food waste tips straight into the sealed hopper; the integral shredder prepares it in a single pass. Limited pre-sorting beyond removing plastics.

02 Compost in 24 hours

An insulated SS304 trough holds 55–65 °C with microorganism dosing; a humidity sensor ends the cycle the moment the batch is done.

03 Air leaves clean

Exhaust gives up its heat to the incoming air, drops its condensate, then passes two activated-carbon beds before the vent.

04 Compost out

80–90% volume reduction. Output cures on racks into dark, soil-like compost for the society's own gardens.

WHY THIS MACHINE

Composting machines fail on three things. We engineered all three out.

Odourcomplaints, electricity billsand corroded chambers are why installed machines fall silent within a fewyears. The DRAVYA treats them as design requirements: a sealed vessel under suction with four stages of exhaust treatment, solar and recovered heat in place of grid heating, and SS 304 on every surface the compost touches.



Side elevation · hopper to vent, everything in one envelope

VESSEL

SS 304 U-trough, Umiporous insulation

Stainless steel on every wettedsurface;mineral-wool insulation to IS 8183 on shell and doors.

HEAT RECOVERY

Counter-flow recuperator

Hot,saturated exhaust preheatsthe incoming air and sheds condensate — cutting heater duty by roughly a third.

ODOUR

Four stages to the vent

Sealed vessel under-suction, condensate knock-out, then twin activated-carbon beds. Odour-free at the property boundary.

CONTROL

Humidity endpoint, PID heat

PLC with touchscreen, the heater stops when the batch reaches target moisture, not when a timer guesses.

SAFETY

Interlocked by design

Blades stop when the lid opens. Emergency stop, overload relays, heater thermal cutouts.

INSIDE THE MACHINE

ENERGY Engineering where it cannot be seen

The section below isthe machine as built:insulation thickenoughto matter, heat exchanged instead of vented, and air kept below room pressure so odour has no way out.



Sectioned view · blue: incoming air, amber: hot exhaust through the recuperator and carbon beds

HYBRID ENERGY

The sun pays the largest bill

Most of a composting machine's electricity goes into evaporating water. The DRAVYA takes that heat from an evacuated-tube solar bank first up to 40% of annual heater demand on a Mumbai roof with the grid heater as night and monsoon trim. Together with heat recovery, grid draw falls to roughly half of a conventional 24-hour machine.

Up to 40%
of annual heating demand from the solar bank

~½ grid draw
versusaconventional24-hour machine

~₹1.9 lakh
a year saved at 500 kg/day*



Solar thermal bank, ~1.2 m^2 per 100 kg/day of capacity

IN SERVICE

Five minutes a day, one bin at a time

House keeping staff feed it like a bin lid up, tip, lid down. The machine does the rest and messages the service team if anything drifts. It stands on 3 m² of podium, yard or basement.



Daily feeding through the interlocked hopper

Installed footprint from 1.4 × 1.0 m



Interlocked lid
Mixingstopstheinstant the lid lifts.

One screen
Temperatures, humidity, cycle state and alerts mirrored to a remote dashboard.

Carbon in minutes
Filtercartridgesswapby hand, no tools, twice a year.

THE RANGE

Seven sizes, one architecture



Model	Capacity	Footprint (L×W×H, m)	Connected load	Energy/day*	With solar*
DRAVYA 100	100 kg/day	1.4 × 1.0 × 1.4	6 kW	~16 kWh	~12 kWh
DRAVYA 250	250 kg/day	1.8 × 1.1 × 1.5	10 kW	~40 kWh	~29 kWh
DRAVYA 500	500 kg/day	2.2 × 1.3 × 1.6	18 kW	~80 kWh	~58 kWh
DRAVYA 750	750 kg/day	2.6 × 1.4 × 1.7	24 kW	~120 kWh	~87 kWh
DRAVYA 1000	1,000 kg/day	3.0 × 1.5 × 1.8	30 kW	~160 kWh	~116 kWh

All models: 24 h cycle + SS 304 wetted parts · integral shredder · recuperator · 4-stage odour train · PLC + HMI with remote telemetry · 3- phase, 50 Hz (60 Hz on request). *Design targets at rated load, annual-average solar on a Mumbai profile; validated at factory acceptance testing. 150 and 350 kg/day built to order.

ASSURANCE

Backed the way infrastructure should be

Witnessed acceptance test

Everymachine runsa 72-hourfactory trial at rated load before dispatch; you or your consultant are welcome on the floor.

Performance in writing

Volume reduction,energyper day and boundary odour are contract clauses, not brochure adjectives.

O&M from the makers

Operation and maintenance contracts with consumables, carbon changes and an annual audit — from the team that runs treatment plants every day.

Compliance support

SWM Rules 2016 documentation for bulk waste generators, and support for IGBC / GRIHA credits. CE-path electrical design for export markets.

FLOROSENSE · ECOVRY

ECOVRY is FLOROSENSE's environmental-infrastructure vertical: sewage treatment, waste-to-resource systems and online environmental monitoring, engineered and serviced in India. The DRAVYA carries that operating experience into a single machine.

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Customisation of specifications are subject acceptability of FLOROSENSE Team. Performance figures are engineering design targets at the stated reference conditions (±20%), to be validated at factory acceptance testing, and do not constitute a quotation or guarantee. DRAVYA, SURYAVERT are working names under evaluation. © FLOROSENSE, 2026.